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Research Paper

Illicit cigarette trade in the greater metropolitan area of Costa Rica: a smokers' survey

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ABSTRACT

Background: Illicit cigarette markets are shaped as much by supply-side and enforcement factors as by consumer affordability. In Central America, a subregion that includes major transhipment points such as the Panama Canal, independent measurement of these structural features has been limited, and public debate has been shaped largely by industry-funded estimates.
Methods: We conducted a cross-sectional survey of 613 adult smokers in Costa Rica's Greater Metropolitan Area (November–December 2023) using a geographically stratified intercept non-probabilistic design with sex–age quotas. We classified each respondent's most recent purchase as licit or illicit through photographic pack inspection against a regulatory checklist and a pre-specified price-based rule for single sticks. We estimated penetration of illicit cigarettes in total consumption and prevalence of illicit purchasing.
Results: Illicit cigarettes accounted for 30.5% (95% CI 25.3–35.7) of total consumption; 20.3% of smokers reported an illicit most recent purchase. Pack-based purchases drove the estimate (27.5%); single sticks contributed 3.05%. Findings were relatively stable under price-threshold variation (29.8% at USD 0.12; 36.9% at USD 0.21). The illicit segment was concentrated (Herfindahl–Hirschman Index 0.306 vs 0.172 for the total market), dominated by three unregistered brands, with packs traced primarily to China and South Korea.
Conclusions: Illicit cigarette trade in the Costa Rican GMA is pack-driven and supply-concentrated. The pattern is consistent with organised transhipment supply chains rather than affordability-driven single-stick sales, supporting customs oversight and full implementation of the Protocol to Eliminate Illicit Trade in Tobacco Products.

Introduction

Illicit cigarette markets are shaped by supply-side and enforcement factors — exploitation of free-trade zones, transhipment through third-country hubs, and gaps in customs oversight — as much as by consumer affordability. (Joossens & Raw, 2012) In Central America, a subregion with critical points for the potential entry of illicit trade (e.g. the Panama Canal), independent measurement of these structural features has been limited.

Costa Rica ratified the WHO Framework Convention on Tobacco Control (FCTC) in 2008 and the Protocol to Eliminate Illicit Trade in Tobacco Products (PEIT) in 2016 (World Health Organization, 2016). However, key Protocol provisions, including a track-and-trace system, supply-chain licensing, and strengthened customs controls in free-trade zones and ports, remain pending (Pan American Health Organization, 2021b). Despite a comprehensive tobacco-control law (Law No 9028) and a mixed excise structure introduced in 2012, adult tobacco use prevalence declined only modestly between 2015 and 2022, from 9.1% to 8.7%, and cigarette affordability did not improve over the period: in the 2022 Tobacconomics Cigarette Tax Scorecard, Costa Rica received 0 of 5 points on the affordability-change component (Drope et al., 2024; GTSS Academy, 2024).

The tobacco industry has historically interfered with tobacco control measures across Latin America, including Costa Rica (Drope et al., 2022; Global Tobacco Index, 2025; World Health Organization, 2009). Common strategies include lobbying through front groups, exaggerating the tobacco sector's economic relevance, and disseminating industry-funded reports on illicit trade that rely on opaque methodologies and produce

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figures consistently higher than independent measurements (Gallagher et al., 2019; KPMG, 2025). As a result, industry narratives have often shaped public debate rather than independent, transparent evidence (Paraje et al., 2022). To our knowledge, no independent study has been published for Costa Rica that estimates the illicit cigarette trade using transparent and reproducible methods. This combination of incomplete PEIT implementation, stagnant affordability reductions, persistent industry-shaped narratives, and the absence of independent country-level evidence motivates the present study.

We address this gap by producing the first independent estimate of illicit cigarette trade in the Costa Rica Greater Metropolitan Area (GMA), where more than 80% of the country's smokers live. Combining a survey of adult smokers with photographic pack inspection and a pre-specified price-based rule for single sticks, we estimate the share of illicit cigarettes in total consumption, the prevalence of illicit purchasing among smokers, and the brands, countries of origin, and retail channels associated with both the licit and illicit segments. This study has two key features. First, we develop and apply a transparent, price-based classification rule for single-stick purchases, a purchase modality present in retail in Central America (Arevalo et al., 2016; de Ojeda et al., 2013). According to the GATS 2022, 14.6% of smokers reported that their last cigarette purchase was made as single cigarettes. This has been used in other studies in the region (Iglesias et al., 2016; Szklo et al., 2020). Second, by weighting observations by reported smoking intensity, we separate the consumption-weighted penetration of illicit cigarettes from the prevalence of illicit purchasing among individuals, two quantities that are often conflated but carry different fiscal and enforcement implications. The difference between prevalence and penetration of illicit cigarettes has also been highlighted in other studies in the region (Maldonado et al., 2019, 2018; Paraje et al., 2018; Saenz de et al., 2021).

Costa Rica's geographic and structural features, proximity to Panama's transhipment infrastructure, the presence of free-trade zones, porous land borders, and a retail landscape dominated by informal and semi-formal outlets, are shared with much of Latin America. Evidence from this setting, therefore, speaks beyond Costa Rica; our methods offer a transparent, reproducible approach for other Central American countries where independent evidence on illicit trade remains limited.

Materials and methods

Sample design and sample size

The study population consists of smokers residing in Costa Rica's Greater Metropolitan Area (GMA), which spans the provinces of San José, Alajuela, Cartago, and Heredia. The GMA is the country's largest urban agglomeration and is home to 74% of Costa Rica's population (Instituto Nacional de Estadísticas y Censos de Costa Rica, 2018). Based on analysis of GATS Costa Rica 2022 microdata, we estimated that 82.2% (95% CI 78.6% – 85.2%) of current smokers reside in these four provinces, which motivates focusing the study on the GMA as the setting that captures the bulk of the national smoking population (World Health Organization, 2022).

We used a geographically stratified intercept non-probabilistic design with quota sampling by sex and age within provinces. Because smokers do not form a practical household-based sampling frame, a probability sample of smokers was not feasible. Instead, we used density-based spatial clustering of high-traffic venues to achieve broad geographic coverage of the GMA (described below), and recruitment within selected zones was guided by sex–age quotas designed to reproduce the demographic distribution of current smokers reported in GATS Costa Rica 2015, the most recent national survey available when fieldwork was designed (World Health Organization, 2015).

The target sample size was set to achieve adequate precision for the study's primary estimand: the share of cigarettes consumed in the GMA that are illicit. Prior independent estimates of illicit cigarette market share in Latin America have ranged around 20–30% (Iglesias et al., 2016; Maldonado et al., 2025; Paraje et al., 2018; Pizarro et al., 2021), with industry-funded figures for Central America typically well above 30% (KPMG, 2025). Even when the design is non-probabilistic, a heuristic benchmark to estimate the sample size was estimated using the standard simple-random-sample formula for a proportion, with an a priori assumption of illicit share in the range of 20–30%:

$$n = \frac{Z^2 * p * (1-p)}{e^2}$$

with $Z = 1.96$ (95% confidence), $p = 0.25$ (midpoint of the assumed range), and $e = 0.04$ (margin of error). This calculation yields a minimum of approximately 450 respondents, which we rounded upward to a target of 600 to provide a buffer for design effects associated with geographic clustering. After cleaning the data, we interviewed 613 adult smokers. Data cleaning involved removing records without a valid survey identification number used to pilot the questionnaire (n = 72); excluding observations with inconsistencies between the reported purchase modality and pack-verification information collected during the interview (n = 23 of which n = 20 reported buying a pack but did not show it); removing observations with no information on smoking intensity (n = 5); and deleting observations reporting average cigarette consumption above 40 cigarettes per day, which were identified as implausible outliers and excluded from the analytical sample (n = 8) (see Fig. S1 in the Appendix for a diagram detailing these steps).

We distributed the total sample across provinces according to their relative share of the GMA population. Within each province, we set quotas by sex and age group to ensure the sample mirrored the known demographic distribution of smokers, as estimated by GATS and applied to province-level adult population figures from INEC (Instituto Nacional de Estadísticas y Censos de Costa Rica, 2018; World Health Organization, 2022). This representativeness stratification aimed to preserve representativeness along key sociodemographic dimensions.

Because no sampling frame of smokers exists at the sub-cantonal level in Costa Rica, we used geographic stratification of the GMA to guide fieldwork. Using Geographic Information System (GIS) tools, we built a geodatabase of population attractors — venues associated with high pedestrian circulation — drawn from Google Maps. These included retail establishments (shops, restaurants, bars), workplaces, educational institutions, parks, public transportation terminals, and shopping centres. We excluded residential areas without these attractors. Alternative approaches, such as household-based recruitment with smoker screening, were not feasible within the project's resources. The intercept design was also chosen because estimating the share of consumed cigarettes that are illicit, our main goal, depends on observing the most recent purchase, with the respondent ideally carrying the pack at the time of interview. This design is consistent with recent independent illicit-trade measurement in the region, which has similarly worked outside a household-based frame (Gallego et al., 2020; Maldonado et al., 2018; Paraje et al., 2018).

We identified spatial clusters of population attractors using hierarchical density-based clustering (HDBSCAN), which groups points by local density without requiring a pre-specified number of clusters (Campello et al., 2013). This procedure was applied sequentially across four nested spatial scales: province, canton, district, and localised zone. At the provincial level, we identified a primary attractor cluster within each province as the largest high-density cluster generated by the HDBSCAN algorithm. Subsequent clustering at the cantonal and district levels progressively refined these areas to smaller, higher-density subclusters. At the district level, we enumerated clusters containing at least 500 attractor entities and retained a random subset for further analysis. In the final stage, we identified localised clusters with at least 200 entities and randomly selected a subset as fieldwork zones. The final set of zones was allocated across provinces in proportion to each province's share of the GMA adult population, yielding 24 fieldwork zones distributed across the four provinces of the GMA (see Appendix

Table S1). This procedure was designed to maximise geographic coverage of high-traffic population areas while maintaining the provincial sampling outputs. Fig. 1 shows the geographic distribution of the resulting fieldwork observations.

We conducted fieldwork between 25 November and 15 December 2023. Within each zone, university-trained interviewers approached adults in public spaces and screened them for eligibility (≥18 years old, current smoker, GMA resident). Eligible respondents who consented were interviewed face-to-face using a structured questionnaire administered on tablets. Recruitment within each zone continued until the pre-specified sex-age quotas, derived from the distribution of current smokers in GATS Costa Rica 2015, were met (World Health Organization, 2015). Respondents who declined to participate or were unwilling or unable to display a cigarette pack for the pack-inspection module were excluded from the analytical sample.

Supervisors monitored all fieldwork in real time via the electronic data-capture platform and conducted unannounced spot checks in each zone to verify adherence to protocol, respondent eligibility, and pack-photography quality. Interviewers received three days of training before deployment, including instruction on the informed-consent procedure, pack-photography protocol, and handling refusals.

The Scientific Committee of Universidad Nacional de Costa Rica approved the study protocol, including the informed consent procedure and data protection measures for pack photographs (approval no UNA-CECUNA-OFIC-109-2023). All respondents provided verbal informed consent before participating, and we anonymised and analysed all survey responses and photographic material in aggregated form.

Data collection and classification of observations

We interviewed adult smokers face-to-face using a structured questionnaire administered on tablets. The instrument collected socio-demographic information (sex, age, province of residence, educational attainment, employment status), smoking behaviour (average daily consumption, age of smoking onset, stated brand and flavour preferences), and details of the respondent's most recent cigarette purchase, such as purchase modality (pack or single stick), reported price, brand, and point of sale. Respondents who reported that their most recent purchase was a pack and were carrying it at the time of the interview were asked to display it for photographic documentation. Enumerators photographed all visible pack surfaces (front, back, both lateral sides, top, and bottom), generating the visual material used in pack classification. Respondents who were unwilling or unable to display a pack were excluded from the pack-classification analysis.

Each photographed pack was reviewed by at least one member of the research team against a pre-specified regulatory checklist derived from Costa Rican tobacco control legislation (Law No 9028) and Ministry of Health regulations in force during fieldwork. A pack was classified as licit only if it met all six criteria: (i) the brand was registered with the Ministry of Health; (ii) the pack complied with import-registration

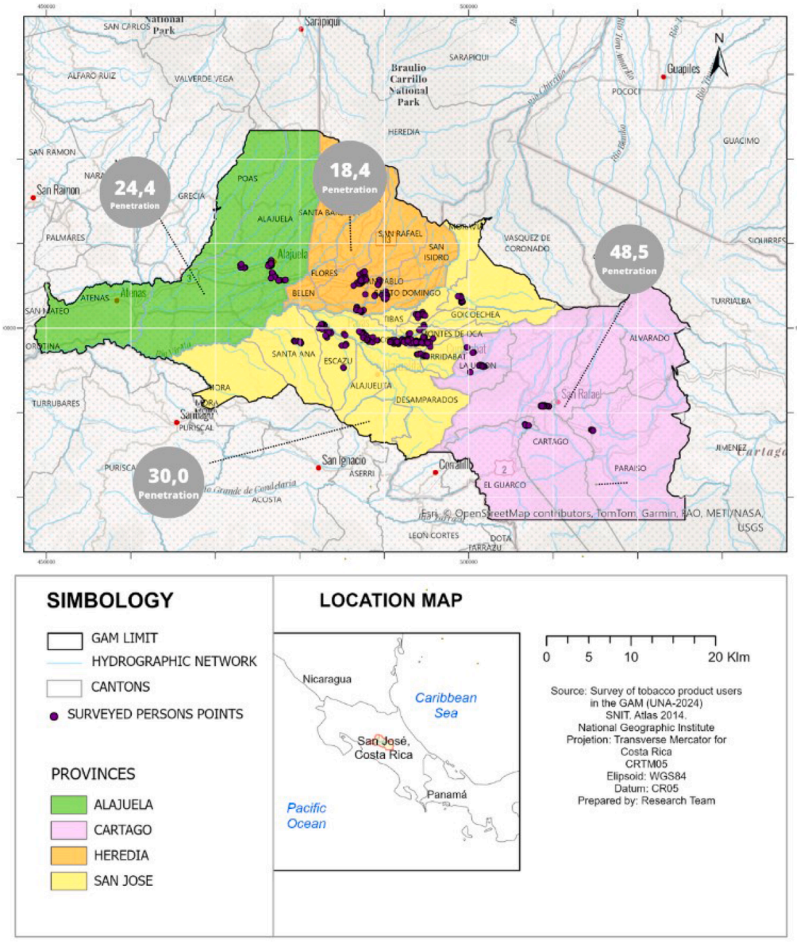


Fig. 1. Geographic distribution of fieldwork observations and province-level penetration of illicit cigarette consumption, Greater Metropolitan Area, Costa Rica, 2023.

Note. Points represent completed interviews (n = 613). Provincial percentages are consumption-weighted illicit-share estimates. Source: 2023 Survey on Illicit Cigarette Trade in the Greater Metropolitan Area. Non-probability intercept design with sex–age quotas. Intervals are descriptive measures of sampling variability and should not be interpreted as design-based confidence intervals.

requirements; (iii) the pack contained the legally required number of cigarettes (20 per pack); (iv) it displayed the mandatory text indicating exclusive sale in Costa Rica and prohibition of sale to minors; (v) it complied with the official colour and design specifications; and (vi) it carried one of the pictorial health warnings authorised for the labelling campaign in force during the study period. Packs failing any criterion were classified as illicit.

Because individual cigarettes did not permit inspection of packaging features, we could not apply pack-based classification. For respondents whose most recent purchase was a single stick, we classified the transaction as illicit using a price-based rule, on the rationale that sticks sold below the minimum threshold price compatible with tax-compliant retail cannot correspond to a licit product, as done elsewhere (Iglesias et al., 2016). Selling single sticks in Costa Rica is forbidden. In the stricter sense of the word, all single sticks are illegal. However, we consider them illicit if, after using the price-based rule, we believe they have not paid national tobacco taxes.

To construct the threshold price, we decomposed the retail price of the most consumed licit brand in our sample (Pall Mall), which sold for approximately CRC 2100 per pack of 20 cigarettes during the fieldwork period (about USD 0.19 per stick). From this retail price, we subtracted the four applicable tax components in force in 2023: the specific excise tax (CRC 24.51 per stick, adjusted annually by CPI), the ad valorem excise tax, value-added tax (VAT), and the INDER contribution. The residual, net-of-tax value corresponded to approximately USD 0.084 per stick (approximately CRC 45 at the fieldwork-period exchange rate). This value represents the theoretical minimum at which a Pall Mall stick could be sold without paying any taxes. For the cheapest licit brand in our sample, the analogous net-of-tax value was approximately USD 0.057 per stick. In markets where single-stick sales are widespread, the literature suggests that per-stick retail prices for licit product typically exceed the pack-per-stick equivalent because retailers capture an additional margin on fragmented sales (Maldonado et al., 2019; Saenz de et al., 2021). However, the magnitude of this markup is context-specific and has not been documented for Costa Rica.

In any case, single-stick licit cigarettes would be unlikely to sell below the 0.06–0.09 price interval, which represents the net-of-tax minimum. We chose USD 0.19 as the baseline threshold (more than double the theoretical minimum), reflecting the pack-per-stick equivalent for the most-sold licit brand. If licit sticks are sold below (above) the baseline threshold, there would be an overestimation (underestimation) of single-stick illicit penetration. To bracket this uncertainty, we pre-specified two sensitivity thresholds: USD 0.12, closer to the net-of-tax minimum viable price; and USD 0.21, above the licit pack-per-stick equivalent. Appendix Figure S2 presents the empirical distribution of self-reported unit prices for single-stick purchases in our sample.

For pack purchases, the coding team read brand and country of origin directly from the photographed pack; for single-stick purchases, they used respondent self-report and, where possible, cross-validated against the retailer's surrounding stock. Respondents self-reported the point of sale, which we coded into a pre-specified taxonomy (supermarket, corner store [*pulpería*], street vendor, liquor store, other). The survey collected information only on the respondent's most recent cigarette purchase and did not seek to quantify the proportion of purchases obtained from different retail sources or the share of total cigarette consumption accounted for by licit versus illicit products. The study therefore characterises the licit or illicit status of the most recent transaction rather than the respondent's entire smoking history or purchasing portfolio.

Statistical analysis

We estimated two quantities: (i) the penetration of illicit cigarettes in total cigarette consumption, defined as the share of cigarette sticks consumed that were classified as illicit; and (ii) the prevalence of illicit purchasing among smokers, defined as the share of respondents who reported an illicit purchase. The two quantities differ because heavier smokers contribute disproportionately to total consumption; we report both because they carry different policy implications — penetration for fiscal impact and enforcement prioritisation, prevalence for measuring exposure at the individual level.

To estimate penetration, we converted each observation into a cigarette-unit equivalent by multiplying by the respondent's self-reported average daily consumption. The penetration of the illicit cigarette trade (PICT) was then calculated as:

$$PICT = \frac{\sum_{k=1}^{I} q_k}{\sum_{j=1}^{T} q_j} \, x \, 100$$

where T is the total number of cigarette observations (pack and stick), I is the subset classified as illicit, and q is the corresponding cigarette-unit equivalent. Prevalence was computed as the unweighted share of respondents reporting at least one illicit purchase. We tabulated brand-level market shares for the overall and illicit segments and computed the Herfindahl–Hirschman Index (HHI) to characterise market concentration.

Confidence intervals for proportions were computed using logit-transformed estimators to ensure that bounds remain within the unit interval. Because the design combines probability selection of zones with quota-based intercept recruitment within zones, intervals should be interpreted as descriptive measures of statistical precision rather than as design-based estimates with strict probability-sampling guarantees.

We pre-specified two sensitivity analyses to assess the robustness of the headline penetration estimate to the classification rule. First, we varied the single-stick price threshold to USD 0.12 (lower bound, closer to the net-of-tax minimum) and USD 0.21 (upper bound, above the licit pack-per-stick equivalent). Second, we computed pack-only and stick-only penetration estimates separately to isolate each purchase modality's contribution.

Finally, to characterise the statistical association with individual-level illicit cigarette purchasing, we fitted a probit model with a binary outcome indicating whether a respondent's most recent purchase was classified as illicit. Covariates included demographic characteristics (sex, age, education, employment status, province of residence), smoking behaviour (age of initiation, daily consumption, stated price and flavour preferences), point of sale, and the logarithm of the cigarette unit self-reported price. Because unit self-reported prices are used to classify single sticks as licit or illicit, this variable cannot be used as a covariate in the case of respondents purchasing single sticks. Hence, we estimated the probit model only using data from respondents purchasing packs. We emphasise that these estimates are descriptive associations and should not be interpreted causally. We conducted all analyses in Stata 18.

Results

Fieldwork yielded 613 observations that met the eligibility criteria (aged ≥18, current smoker, GMA resident). Of these, 355 respondents whose most recent purchase was a pack were able to display the pack for photographic inspection. Fig. S1 in the Appendix presents the full sample-flow diagram. The realised sample exceeded the pre-specified target of 600 and reproduced the intended geographic allocation: fieldwork was organised into 24 zones across the four provinces of the GMA, with zone allocation reflecting each province's share of the GMA adult population. San José had the most zones, followed by Heredia, Cartago, and Alajuela (Appendix Table S1 and Fig. 1).

The final sample also broadly approximated the sex and age distribution of current smokers reported in GATS Costa Rica 2015: 75.2% male (GATS 2015: 75.4%) and 47.3% aged 36–64 (Appendix Table S2). Educational attainment was skewed toward lower levels, with 44.8% of respondents reporting primary or incomplete secondary education, and more than two-thirds were employed at the time of the interview.

Average daily consumption was 8.7 cigarettes (95% CI: 8.0–9.3), confirming the substantive importance of intensity weighting for the penetration estimate. 57.7% of respondents reported that their most recent purchase was a pack, and 42.3% that it was a single stick — a purchase-modality split that motivates the separate treatment of pack inspection and the price-based rule in the Methods. The composition of our sample differs from GATS 2015 (and GATS 2022), as shown in Appendix Table S2. This difference likely reflects the different sample design: our intercept design interviews individuals in high-footfall public spaces and may oversample smokers active in informal retail environments where single-stick sales concentrate.

Applying the combined classification — pack inspection for packs and the baseline USD 0.19 per-stick rule for single sticks — we estimate that illicit cigarettes accounted for 30.5% of total cigarette sticks consumed in the GMA in 2023 (95% CI 25.3–35.7). The corresponding prevalence of illicit purchasing, measured as the share of respondents whose most recent purchase was classified as illicit, was 20.3% (95% CI 17.3–23.7). The gap between the two quantities — roughly 10 percentage points — reflects the disproportionate contribution of heavier smokers to total cigarette consumption and illustrates why the two measures are reported separately: penetration is the relevant quantity for fiscal impact and enforcement prioritisation, whereas prevalence captures individual-level exposure.

Table 1 presents the baseline estimate together with the two prespecified sensitivity analyses (by price levels for single sticks). Disaggregating by purchase modality, pack-based illicit consumption accounted for 27.5% (95% CI 22.3–32.6) of total cigarette sticks, while illicit single-stick purchases accounted for only 3.05% (95% CI 1.4–4.6). The pack-based component, therefore, drives the headline estimate. This pattern is consistent across the price-threshold sensitivity analyses: at a lower threshold of USD 0.12, total penetration was 29.8% (95% CI 24.6–35.1); at a higher threshold of USD 0.21, total penetration was 36.9% (95% CI 31.5–42.3). The total penetration estimate is thus relatively stable across alternative classification rules for single sticks.

Fig. 1 displays the geographic distribution of fieldwork observations across the GMA, along with province-level estimates of the illicit trade cigarette penetration. Penetration varied markedly across the GMA's four provinces. Cartago showed the highest estimated penetration at 48.5% (95% CI 35.4–61.5), followed by San José at 30.0% (95% CI 22.6–37.4) and Alajuela at 24.4% (95% CI 11.1–37.7), while Heredia showed the lowest at 18.4% (95% CI 9.0–27.8). The wide confidence intervals, particularly for Cartago and Alajuela, reflect both the smaller number of zones and the smaller cell sizes at the provincial level; provincial estimates should therefore be read as indicative rather than definitive, and we do not draw pairwise comparisons between provinces.

Brand information was available from both pack purchases (read from the photographed pack) and single-stick purchases (respondent self-report); both modalities are included in the brand analysis (Table 2). The GMA cigarette market is dominated by internationally distributed, legally registered brands, with four legal brands accounting for more than half of total consumption: Pall Mall (30.5%), Chesterfield (13.1%), Marlboro (12.1%), and L&M (9.9%). Three brands not registered with the Ministry of Health together account for 26.7% of total cigarette consumption in the GMA: Gold City (17.6%), Elephant (5.5%), and Indigo (3.6%). Gold City alone is the second-largest brand overall, ahead of every legal brand except Pall Mall. The substantial presence of unregistered brands in the overall consumption mix, not just at its margins, is the first indication that illicit cigarettes circulate through mainstream rather than peripheral retail channels in the GMA. Brands owned by British American Tobacco (BAT) and Philip Morris International (PMI) dominated the legal segment, accounting for most legal cigarette consumption observed in the GMA. By contrast, the illicit segment was concentrated in a small number of unregistered brands that are not affiliated with major transnational tobacco companies.

Restricting attention to the illicit segment reveals a highly concentrated brand structure (Table 2). Gold City alone accounts for 50.3% of illicit cigarette consumption (95% CI, 39.5–61.0), followed by Elephant (17.0%) and Indigo (9.1%). Together, these three brands account for more than three-quarters of illicit consumption in the GMA. The remainder of the illicit segment is composed of (i) smaller unregistered brands (Gold Mine, Ultra Buy Menthol King Box, Royal Pacific Finest Tobaccos, and others, each below 2%) and (ii) legally registered brands appearing in non-compliant form — most notably L&M (5.1%), Pall Mall (3.4%), Chesterfield (2.2%) and Marlboro (2.0%).

The presence of legally registered brands such as L&M, Chesterfield and Marlboro within the illicit segment does not imply counterfeit production. Rather, these cases reflect regulatory non-compliance: packs with missing or incorrect health warnings, unauthorised packaging characteristics, or — for single sticks — retail prices below the operational threshold. The pattern suggests that the illicit market in the GMA

Table 1
Illicit cigarette market share in the GMA, 2023: baseline and sensitivity analyses. Total sample of 613 observations.

Classification	Illicit share (%)	95% CI
Baseline	**30.5**	25.3–35.7
Pack-only	27.5	22.3–32.6
Stick-only	3.05	1.4–4.6
Sensitivity (USD 0.12)	29.8	24.6–35.1
Sensitivity (USD 0.21)	36.9	31.5–42.3
Prevalence of illicit purchasing among smokers	20.3	17.3–23.7

Source: 2023 Survey on Illicit Cigarette Trade in the Greater Metropolitan Area. 95% confidence intervals were computed using logit-transformed proportions to ensure that the bounds were within 0–100%. Non-probability intercept design with sex–age quotas. Intervals are descriptive measures of sampling variability and should not be interpreted as design-based confidence intervals. Sensitivity analyses vary the price threshold used to classify single sticks only; pack-based classifications are unchanged. The reported 'total' figures reflect the recomposition of pack and stick components under each threshold.

Table 2
Cigarette brands and market share in total and illicit segments, GMA, 2023. Total sample of 613 observations.

Brand	Total market % (95% CI)	Illicit market % (95% CI)	Legal status	Company owner
Pall Mall	30.5 (25.7–35.7)	3.4 (1.18–9.83)	*Registered*	*BAT*
Gold City	17.6 (13.5–22.7)	50.3 (39.5–61.0)	*Not registered*	-
Chesterfield	13.1 (9.7–17.4)	2.2 (0.6–7.8)	*Registered*	*PMI*
Marlboro	12.1 (9.1–16.0)	2.0 (0.5–7.4)	*Registered*	*PMI*
L&M	9.9 (7.2–13.4)	5.1 (2.0–12.1)	*Registered*	*PMI*
Elephant	5.5 (3.3–9.1)	17.0 (10.1–27.2)	*Not registered*	-
Indigo	3.6 (1.9–6.7)	9.1 (4.8–16.3)	*Not registered*	-
Derby	2.5 (1.5–4.2)	-	*Registered*	*PMI*
Others	4.8	10.7	*Mixed*	

Source: 2023 Survey on Illicit Cigarette Trade in the Greater Metropolitan Area. Brand information was obtained from two sources. For pack purchases, brands were recorded directly from photographed packs inspected during the interview. For single-cigarette purchases, brands were obtained from respondent self-report. In "Total market", the category "Others" includes brands with individual market share below 1%: Seneca, Gold Mine, Ultra Buy Menthol King Box, Altima, Royal Pacific Finest Tobaccos, Lucky Strike, Fisher, Bison, Camel, Couture, Landus Red, Rothmans, Kool, and Hul. BAT is British American Tobacco; PMI is Philip Morris International. 95% confidence intervals were computed using logit-transformed proportions to ensure bounds remain within 0–100%. Non-probability intercept design with sex–age quotas. Intervals are descriptive measures of sampling variability and should not be interpreted as design-based confidence intervals. Legal status refers to the registration status of each brand with the Costa Rican Ministry of Health. Brands not registered for sale in Costa Rica were classified as illicit. Registered brands appearing in the illicit segment correspond to non-compliant products that failed at least one element of the regulatory checklist used in the study.

is not confined to clearly illicit brands but also involves diverting recognised legal products into non-compliant retail channels.

To quantify the structural difference between the total and the illicit cigarette markets, we computed the Herfindahl–Hirschman Index (HHI) on brand-level market shares for each segment (full values in Appendix Table S3). On the 0–1 scale, HHI values below 0.15 are conventionally described as unconcentrated, 0.15–0.25 as moderately concentrated, and above 0.25 as concentrated (Department of Justice, Federal Trade Commision, 2010). The total GMA cigarette market (licit and illicit) yielded an HHI of 0.172, consistent with a moderately concentrated market dominated by multiple international brands. The illicit segment, by contrast, yielded an HHI of 0.306 — almost doubling the overall value and above the conventional threshold for a concentrated market — reflecting the dominance of Gold City and the narrow set of brands in the illicit segment.

Country-of-origin information was obtained from pack labelling for the subset of illicit packs with legible origin markings. Among the 100 illicit packs identified in the sample, origin information could be classified for 37 packs (37%). Within this subset, China accounted for 42.3% of identifiable illicit packs and South Korea for 41.3%, together representing more than four-fifths of the observable sample (Table 3). Because this analysis excludes single sticks (for which origin cannot be gauged) and illicit packs with obscured or absent origin labelling, these shares describe the observable composition of illicit packs rather than the full illicit market.

Points of sale differ markedly by licit status and by purchase modality (Table 4). Licit pack purchases were concentrated in supermarkets (53.0%) and corner stores (38.0%), with negligible shares from street vendors. Illicit pack purchases showed an almost reversed pattern: corner stores accounted for more than half (51.9%) of illicit pack purchases, street vendors for a further 17.2% — compared to 0% for licit pack purchases — and supermarkets for only 17.1%. Supermarkets therefore account for a threefold larger share of licit than of illicit pack purchases, while street vendors account for essentially no licit pack purchases but for one in six illicit pack purchases.

For single sticks, both licit and illicit purchases were concentrated in corner stores (49.9% and 56.4%, respectively) and street vendors (22.8% and 19.8%), with similar distributions across the two categories. Confidence intervals for illicit stick shares are wide, reflecting the smaller cell size and the conservative nature of the price-based classification rule.

Table S4 in the Appendix reports average marginal effects from an exploratory probit model of illicit purchasing among survey respondents purchasing packs. Two individual-level factors emerged as robust correlates. Higher cigarette unit self-reported prices were strongly associated with a lower probability of an illicit purchase (AME −0.25 percentage points per 1% increase in price; $p < 0.001$), and, consistent with this, respondents who stated a preference for lower-priced cigarettes showed a substantially higher probability of illicit purchasing (AME +20.8 pp; $p < 0.001$). Purchase channel also mattered: relative to corner stores (the reference category), purchases made in supermarkets were significantly less likely to be classified as illicit (AME −6.2 pp; $p = 0.042$). A similar pattern was observed for purchases made through "other" retail channels (AME −19.9 pp; $p < 0.001$) which includes bars and restaurants, online and delivery purchases. Liquor stores showed no statistically significant difference. In contrast, sociodemographic characteristics—sex, age, age of smoking initiation, education, employment status, and province of residence—were not statistically associated with illicit purchasing, except for residence in Heredia, which was associated with a lower probability of illicit purchasing. Effects of sociodemographic characteristics were small in magnitude. Discrimination was high (AUC 0.966), but the results should be interpreted as descriptive associations rather than causal effects, given the cross-sectional design and the absence of instrumentation for price.

Table 3
Country of origin of illicit cigarette packs with legible origin information, GMA, 2023.

Country of origin	Share of illicit packs (%)	(95% CI)
China	42.3	(25.8–58.7)
South Korea	41.3	(25.0–57.6)
Brazil	5.3	(0.0–12.7)
United States	5.3	(0.0–12.6)
Switzerland	3.0	(0.0–8.9)
Mexico	2.8	(0.0–8.2)

Source: 2023 Survey on Illicit Cigarette Trade in the Greater Metropolitan Area. Excludes illicit packs without legible origin information and illicit single sticks. 95% confidence intervals were computed using logit-transformed proportions to ensure that the bounds were within 0–100%. Non-probability intercept design with sex–age quotas. Intervals are descriptive measures of sampling variability and should not be interpreted as design-based confidence intervals.

Table 4
Points of sale for licit and illicit cigarette purchases by purchase modality (packs vs single sticks), GMA, 2023. Total sample of 613 observations.

	Packs purchases, % (95% CI)		Single-stick purchases, % (95% CI)	
Point of sale	Licit	Illicit	Licit	Illicit
Corner store	38.0 (32.1–44.0)	51.9 (42.1–61.7)	49.9 (43.5–56.3)	56.4 (37.0–75.8)
Supermarket	53.0 (46.9–59.1)	17.1 (9.7–24.5)	13.3 (8.9–17.6)	11.7 (0.0–24.2)
Street vendor	0.0	17.2 (9.8–24.6)	22.8 (17.4–28.2)	19.8 (4.2–35.3)
Liquor store	5.1 (2.4–7.7)	9.8 (4.0–15.5)	8.8 (5.2–12.4)	0.0
Other	3.9 (1.5–6.2)	4.1 (0.2–8.0)	5.2 (2.3–8.0)	12.1 (0.0–24.9)

Source: 2023 Survey on Illicit Cigarette Trade in the Greater Metropolitan Area. "Other" includes duty-free shops, bars and restaurants, online and delivery purchases, and "don't know/don't remember." 95% confidence intervals were computed using logit-transformed proportions to ensure bounds remain within 0–100%. Non-probability intercept design with sex–age quotas. Intervals are descriptive measures of sampling variability and should not be interpreted as design-based confidence intervals.

Discussion

This study provides the first independent assessment of illicit cigarette market penetration in Costa Rica's GMA, combining photographic pack inspection with an explicit price rule for single-stick purchases. The results indicate that illicit cigarettes account for 30.5% of all cigarette sticks consumed in the GMA. This estimate is considerably lower than industry-funded figures commonly cited in public debate, yet large enough to warrant serious attention from fiscal and public health authorities. A recent KPMG study funded by Philip Morris International states that illicit trade represented 46% of the total national market in 2024. (KPMG, 2025)

A key contribution of this study is the distinction between penetration and prevalence of illicit trade. While 20.3% of smokers reported an illicit most recent purchase, market share estimates weighted by consumption intensity show a higher penetration. This difference highlights the importance of focusing on cigarette flows rather than solely on individual exposure when assessing fiscal losses and enforcement priorities, as heavier smokers contribute disproportionately to total consumption, as found elsewhere (Drope et al., 2022; Maldonado et al., 2019; Paraje et al., 2018; Paraje et al., 2022).

A central policy-relevant finding is that illicit cigarette consumption in the GMA is predominantly associated with full cigarette packs purchased in established retailers rather than single-stick purchases from street vendors. Our data suggest a supply-driven pattern in which unregistered brands are distributed through established retail channels and reach consumers via full packs. Whether the pattern documented here

holds across other Central American settings, and whether it changes under different enforcement regimes, are open empirical questions.

The brand composition and concentration analysis reinforce this interpretation. The illicit segment shows a substantially higher concentration than the overall cigarette market, with an HHI of 0.306 versus 0.172 for the total market (Appendix Table S3). The dominance of a small number of illicit brands, particularly Gold City, is consistent with organised supply chains and recurring sourcing mechanisms, rather than a fragmented network of sporadic illicit actors.

Country-of-origin evidence further supports this interpretation. Country-of-origin information was unavailable for a substantial share of illicit cigarettes because labelling was illegible or absent. Among those with an identifiable origin, China and South Korea together account for more than 80% of illicit packs. While this distribution reflects only the observable subset of the illicit market, it is unlikely to be consistent with local production or small-scale cross-border trade. Instead, the predominance of geographically distant source countries is consistent with the possible role of transhipment and structured long-distance supply routes rather than exclusively opportunistic smuggling across neighbouring borders. This pattern aligns with the high brand concentration observed in the illicit segment and reinforces the interpretation of an organised, supply-driven market rather than a fragmented or purely demand-driven phenomenon. The country-of-origin information printed on the pack is not always correct, as found in other cases. (Joossens & Raw, 2012) Though we have no evidence this is the case for Costa Rica.

The dominance of corner stores and street vendors in illicit pack distribution, combined with the near-absence of street vendors in licit pack distribution, indicates that illicit cigarette distribution is closely tied to informal and semi-formal retail rather than formal large-format outlets. This finding, together with the brand concentration and country-of-origin patterns described above, gives a consistent structural picture of how illicit cigarettes reach consumers in the GMA.

Exploratory regression results (reported in the Appendix, Table S4) suggest that price sensitivity and lower unit prices are associated with a higher likelihood of illicit purchasing, while purchases made in formal retail outlets are associated with lower illicit exposure. These associations are descriptive and should not be interpreted causally; however, they are consistent with the broader evidence that illicit trade reflects supply-side availability and enforcement gaps rather than tax policy alone.

The observed patterns (i.e., pack-driven consumption, concentrated brand structure, etc.) are more consistent with supply-side and enforcement gaps (at the retail level, oversight of transhipment and free trade operations) than with an interpretation of illicit trade as primarily driven by domestic consumer affordability (Chávez Rodríguez, 2023; Krylova, 2024; Organized Crime & Corruption Reporting Project, 2022; Pan American Health Organization, 2021a; Transnational Crime & Corruption Center, 2023; World Customs Organization, 2023; World Health Organization, 2021). The affordability-driven explanation has been mostly advanced by the tobacco industry and claims that illicit cigarettes principally serve smokers who cannot afford licit cigarettes at prevailing prices, and that the primary policy lever to reduce illicit trade is therefore tax restraint or freeze rather than enforcement and supply-chain controls (KPMG, 2025). We note that this argument, while present in policy debate, has been contested by independent, peer-reviewed literature documenting supply-side determinants of illicit trade in Latin America (Joossens & Raw, 2012; Paraje et al., 2022). We emphasise that our cross-sectional design does not allow causal statements about the relative importance of taxation versus enforcement gaps as drivers of illicit trade; the findings should be read as descriptive evidence about the structure of the illicit market, offering hypotheses about mechanisms rather than tests of them.

This study has several limitations worth discussing. First, because smokers do not form a practical household-based sampling frame in Costa Rica, the study relied on a geographically stratified intercept design with sex–age quotas rather than a probability sample. While quota targets closely reproduced the demographic distribution of current smokers in GATS Costa Rica 2015 (Appendix Table S2), the design does not permit strict design-based inference; intervals reported throughout should be interpreted as descriptive measures of precision rather than as intervals with formal probability-sampling guarantees. This approach is consistent with prior independent illicit-trade studies in the region, for which no probability sample of smokers has been feasible either (Maldonado et al., 2025; Paraje et al., 2018).

Second, our single-stick classification rule identifies sticks as illicit if they sell below USD 0.19, a threshold anchored in the pack-per-stick retail price of the most consumed licit brand, more than double its net-of-tax value. The headline penetration estimate is nonetheless stable across the pre-specified price-threshold sensitivity analyses (29.8% at USD 0.12; 36.9% at USD 0.21), and the pack-driven structure of the illicit market — not the single-stick component — carries the interpretive weight of the analysis.

Third, for single-stick purchases, we used respondent self-report for brand, country of origin, and point of sale, since no physical pack was available for inspection. Recall bias and misidentification are therefore possible. Where feasible, self-reported brand and origin were cross-validated against the retailer's surrounding stock, but this validation was not systematic. Point of sale was self-reported for both packs and sticks and coded into a pre-specified taxonomy that may not fully capture hybrid retail formats (for example, corner stores selling both licit and unregistered brands from the same counter).

Fourth, while the sample size was calibrated to achieve adequate precision for the primary estimands (i.e., the share of illicit cigarettes in total consumption across the GMA and its prevalence), precision is inevitably lower for disaggregated sub-estimates. For this reason, provincial estimates should be read as indicative of geographic heterogeneity rather than as definitive province-specific measurements, and we do not draw pairwise comparisons between provinces. Studies aiming to produce reliable province-level estimates, or to compare specific brands or origins within the illicit segment, would benefit from sample sizes calibrated to those sub-estimands specifically.

Fifth, our sample composition differs from GATS Costa Rica 2015 in the share of single-stick purchases (42.3% vs 15.2%). As mentioned in Results, this difference likely reflects the design effect of intercept recruitment in high-footfall public spaces. This implies that our estimates characterise smokers active in public retail environments in the GMA, which may differ from the full smoker population captured by a household-based design. Importantly, however, the study's central interpretive finding (ie. that illicit cigarette consumption in the GMA is pack-driven rather than stick-driven) is unlikely to be affected by this difference.

Finally, we conducted fieldwork over a three-week window in late 2023. Seasonal fluctuations in tobacco pricing, retail behaviour, or enforcement intensity are not captured, and the estimates should be read as a cross-sectional snapshot rather than an annual average. Repeated measurement would allow both temporal validation and assessment of policy impact as PEIT implementation advances. As surveys were collected on the streets, individuals with higher incomes (i.e. higher opportunity cost) may have had a higher refusal rate. If income is related to the choice between licit and illicit cigarettes, this fact could impact our results. However, we have no evidence that this was the case.

Conclusions

This study shows that illicit cigarettes constitute a measurable and policy-relevant share of cigarette consumption in Costa Rica's GMA. Illicit market penetration is unevenly distributed across provinces and characterised by high brand concentration and organised supply chains. The predominance of illicit full-pack consumption highlights supply-chain vulnerabilities that retail enforcement alone cannot address.

The results underscore the importance of completing the implementation of the WHO Protocol to Eliminate Illicit Trade in Tobacco Products, strengthening customs oversight, particularly in free-trade zones and ports, and expanding routine compliance checks at the

retail level. Transparent, reproducible methods such as those employed here are essential to counter industry narratives, inform evidence-based policy, and support effective tobacco control in Costa Rica and the wider region.

Continued independent monitoring is critical. Future research should extend coverage beyond the GMA to border and rural areas and combine pack-based methods with targeted supply–chain audits to better identify upstream points where illicit products enter legal and semi-legal distribution networks.

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Patient and public involvement

Patients and members of the public were not involved in the design, conduct, or reporting of this study. The study collected data from adult smokers via a structured intercept survey and does not involve clinical care, treatment, or patient-facing interventions.

CRediT authorship contribution statement

María Leonela Artavia-Jiménez: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis. **Jorge Solano Ruiz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Formal analysis, Data curation. **Mariela Campos Lizano:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Data curation. **María José Herrera Madrigal:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Formal analysis, Data curation. **Luca Pruzzo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Guillermo Paraje:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationshipsthat could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.drugpo.2026.105543.

Data availability

The anonymised dataset and analysis code are available from the corresponding author on reasonable request, subject to the terms of the study's ethics approval (UNA-CECUNA-OFIC-109-2023). Aggregated data supporting the findings are reported in full in the manuscript and appendix.

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